EXPERIMENT 2

House Price Prediction using Linear Regression

1. Problem Definition

The objective of this experiment is to predict house prices using features such as:

• Location

• Area (sqft)

• Bedrooms

• Bathrooms

• Balcony

• Parking

• Furnishing Status

• Age of the House

Since the output variable (Price) is continuous, this is a Regression Problem.

2. Data Understanding

The dataset contains 120 house records.

Each row represents one house.

Target Variable (y)

• Price

Input Features (X)

Numerical:

• Area_sqft

• Bedrooms

• Bathrooms

• Balcony

• Parking

• Age_Years

Categorical:

• Location

• Furnished

The dataset contains no missing values.

3. Data Preprocessing

(a) One-Hot Encoding

Categorical variables were converted into numerical form using One-Hot Encoding:

• Location_Urban

• Location_Suburban

• Furnished_Yes

This prevents false ordinal relationships between categories.

(b) Feature–Target Split

• X → All features except HouseID and Price

• y → Price

HouseID was removed because it is only an identifier.

4. Train–Test Split

The dataset was divided into:

• 80% Training Set

• 20% Test Set

This ensures model performance is evaluated on unseen data.

Regression Model

Linear Regression was used.

Model form:

$$Price = b0 + b1x1 + b2x2 + ... + bnxn$$

Model Training

The model learned coefficients that minimize Mean Squared Error (MSE) on the training data.

5. Prediction and Evaluation

The trained model predicted house prices for the test dataset.

Mean Squared Error (MSE)

$$MSE = 3.927265130552285 \times 10{-}26$$

This value is extremely close to zero, meaning prediction error is negligible.

R^2 Score

$$R2 = 1.0$$

This means:

- The model explains 100% of the variance in house prices.
- The dataset has a perfect linear relationship.